

```

USE slatask;

CREATE TABLE departments (dept_id INT,dept_name VARCHAR(50));

DROP TABLE employees;

CREATE TABLE employees (emp_id INT,emp_name VARCHAR(50),dept_id
INT,manager_id INT,salary INT);

CREATE TABLE projects (project_id INT,project_name VARCHAR(50),dept_id INT);

CREATE TABLE emp_projects (emp_id INT,project_id INT);

CREATE TABLE locations (location_id INT,dept_id INT,city VARCHAR(50));

INSERT INTO departments VALUES(1, 'IT'),(2, 'HR'),(3, 'Finance'),(4, 'Sales');

INSERT INTO employees VALUES(1, 'Alice', 1, NULL, 80000),(2, 'Bob', 1, 1, 60000),(3,
'Charlie', 2, 1, 50000),(4, 'David', 3, 2, 70000),(5, 'Eva', NULL, 2, 45000);

INSERT INTO projects VALUES(101, 'Website', 1),(102, 'Payroll', 3),(103, 'Recruitment', 2);

INSERT INTO emp_projects VALUES(1, 101),(2, 101),(3, 103),(4, 102);

INSERT INTO locations VALUES(1, 1, 'New York'),(2, 2, 'London'),(3, 3, 'Tokyo');

-- 1. Employees who earn more than the average salary

SELECT *

FROM employees

WHERE salary > (SELECT AVG(salary) FROM employees);

```

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	NULL	80000
	4	David	3	2	70000

```

-- 2. Employees who work in the IT department

SELECT e.*

FROM employees e

JOIN departments d ON e.dept_id = d.dept_id

WHERE d.dept_name = 'IT';

```

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	NULL	80000
	2	Bob	1	1	60000

-- 3. Employees who earn the highest salary

SELECT *

FROM employees

WHERE salary = (SELECT MAX(salary) FROM employees);

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	NULL	80000

-- 4. Employees who do not belong to any department

SELECT *

FROM employees

WHERE dept_id IS NULL;

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	5	Eva	NULL	2	45000

-- 5. Employees working on at least one project

SELECT DISTINCT e.*

FROM employees e

JOIN emp_projects ep ON e.emp_id = ep.emp_id;

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	HULL	80000
	2	Bob	1	1	60000
	3	Charlie	2	1	50000
	4	David	3	2	70000

-- 6. Employees NOT working on any project

SELECT *

FROM employees

WHERE emp_id NOT IN (SELECT emp_id FROM emp_projects);

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	5	Eva	HULL	2	45000

-- 7. Departments that have at least one employee

SELECT DISTINCT d.*

FROM departments d

JOIN employees e ON d.dept_id = e.dept_id;

Output

	dept_id	dept_name
▶	1	IT
	2	HR
	3	Finance

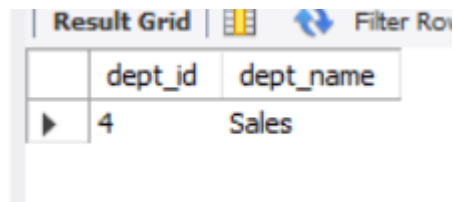
-- 8. Departments with no employees

```
SELECT *
```

```
FROM departments
```

```
WHERE dept_id NOT IN (SELECT dept_id FROM employees WHERE dept_id IS NOT NULL);
```

Output



	dept_id	dept_name
▶	4	Sales

-- 9. Employees who earn more than their manager (Correlated)

```
SELECT e.*
```

```
FROM employees e
```

```
WHERE salary > (
```

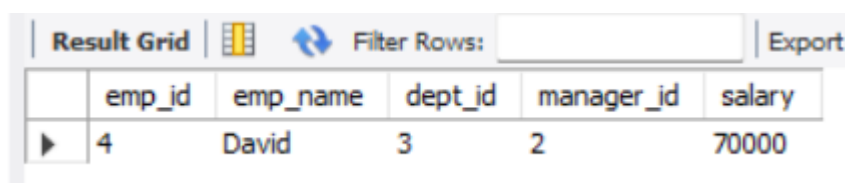
```
    SELECT m.salary
```

```
    FROM employees m
```

```
    WHERE m.emp_id = e.manager_id
```

```
);
```

Output



	emp_id	emp_name	dept_id	manager_id	salary
▶	4	David	3	2	70000

-- 10. Employees who work in departments located in Tokyo

```
SELECT e.*
```

```
FROM employees e
```

```
JOIN locations l ON e.dept_id = l.dept_id
```

```
WHERE l.city = 'Tokyo';
```

Output

Result Grid					
	emp_id	emp_name	dept_id	manager_id	salary
▶	4	David	3	2	70000

-- 11. Employees working on the "Website" project

```
SELECT e.*  
FROM employees e  
JOIN emp_projects ep ON e.emp_id = ep.emp_id  
JOIN projects p ON ep.project_id = p.project_id  
WHERE p.project_name = 'Website';
```

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	NULL	80000
	2	Bob	1	1	60000

-- 12. Projects assigned to employees from IT department

```
SELECT DISTINCT p.*  
FROM projects p  
JOIN emp_projects ep ON p.project_id = ep.project_id  
JOIN employees e ON ep.emp_id = e.emp_id  
JOIN departments d ON e.dept_id = d.dept_id  
WHERE d.dept_name = 'IT';
```

Output

Result Grid			
	project_id	project_name	dept_id
▶	101	Website	1

-- 13. Employees who are managers

SELECT *

FROM employees

WHERE emp_id IN (SELECT DISTINCT manager_id FROM employees WHERE manager_id IS NOT NULL);

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	NULL	80000
	2	Bob	1	1	60000

-- 14. Employees who are NOT managers

SELECT *

FROM employees

WHERE emp_id NOT IN (SELECT DISTINCT manager_id FROM employees WHERE manager_id IS NOT NULL);

Output

Result Grid					
	emp_id	emp_name	dept_id	manager_id	salary
▶	3	Charlie	2	1	50000
	4	David	3	2	70000
	5	Eva	NULL	2	45000

-- 15. Departments that have projects

SELECT DISTINCT d.*

FROM departments d

JOIN projects p ON d.dept_id = p.dept_id

Output

	dept_id	dept_name
▶	1	IT
	2	1
	3	Finance

-- 16. Departments that do not have projects

SELECT *

FROM departments

WHERE dept_id NOT IN (SELECT dept_id FROM projects);

Output

Result Grid		Filter Rows
	dept_id	dept_name
	4	Sales

-- 17. Employees who earn more than the average salary of their department (Correlated)

SELECT e.*

FROM employees e

WHERE salary > (

SELECT AVG(salary)

FROM employees

WHERE dept_id = e.dept_id

);

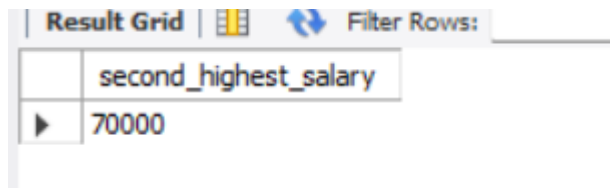
Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	NULL	80000

-- 18. Second highest salary

```
SELECT MAX(salary) AS second_highest_salary  
FROM employees  
WHERE salary < (SELECT MAX(salary) FROM employees);
```

Output



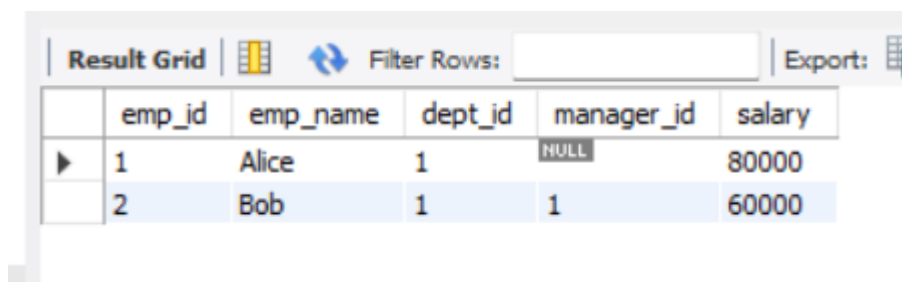
A screenshot of a database application's 'Result Grid'. The grid has two columns: 'second_highest_salary'. The first row contains the value '70000'. Above the grid, there are icons for 'Filter Rows' and 'Export'.

	second_highest_salary
▶	70000

-- 19. Employees working in departments with more than one employee

```
SELECT *  
FROM employees  
WHERE dept_id IN (  
    SELECT dept_id  
    FROM employees  
    GROUP BY dept_id  
    HAVING COUNT(*) > 1  
);
```

Output



A screenshot of a database application's 'Result Grid'. The grid has five columns: 'emp_id', 'emp_name', 'dept_id', 'manager_id', and 'salary'. The first row shows '1', 'Alice', '1', 'NULL', and '80000'. The second row shows '2', 'Bob', '1', '1', and '60000'. Above the grid, there are icons for 'Filter Rows' and 'Export'.

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	NULL	80000
	2	Bob	1	1	60000

-- 20. Employees whose department has a location

```
SELECT e.*
```

```
FROM employees e
```

```
WHERE dept_id IN (SELECT dept_id FROM locations);
```

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	NULL	80000
	2	Bob	1	1	60000
	3	Charlie	2	1	50000
	4	David	3	2	70000

-- 21. Employees whose department does NOT have a location

```
SELECT *
```

```
FROM employees
```

```
WHERE dept_id NOT IN (SELECT dept_id FROM locations)
```

```
OR dept_id IS NULL;
```

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	5	Eva	NULL	2	45000

-- 22. Projects belonging to departments with employees

```
SELECT DISTINCT p.*
```

```
FROM projects p
```

```
JOIN employees e ON p.dept_id = e.dept_id;
```

Output

	project_id	project_name	dept_id
▶	101	Website	1
	103	Recruitment	2
	102	Payroll	3

-- 23. Employees who work on projects from their own department

```
SELECT DISTINCT e.*
```

```
FROM employees e
```

```
JOIN emp_projects ep ON e.emp_id = ep.emp_id
```

```
JOIN projects p ON ep.project_id = p.project_id
```

```
WHERE e.dept_id = p.dept_id;
```

Output

	emp_id	emp_name	dept_id	manager_id	salary
▶	1	Alice	1	NULL	80000
	2	Bob	1	1	60000
	3	Charlie	2	1	50000
	4	David	3	2	70000

-- 24. Employees who work on projects outside their department

```
SELECT DISTINCT e.*
```

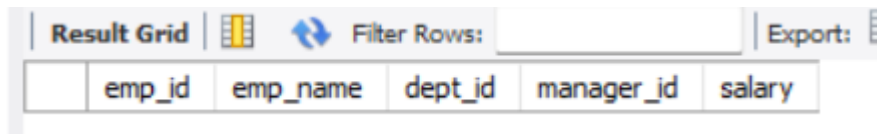
```
FROM employees e
```

```
JOIN emp_projects ep ON e.emp_id = ep.emp_id
```

```
JOIN projects p ON ep.project_id = p.project_id
```

```
WHERE e.dept_id <> p.dept_id;
```

Output



	emp_id	emp_name	dept_id	manager_id	salary
--	--------	----------	---------	------------	--------

-- 25. Departments where all employees earn more than 50,000

SELECT d.*

FROM departments d

WHERE NOT EXISTS (

SELECT 1

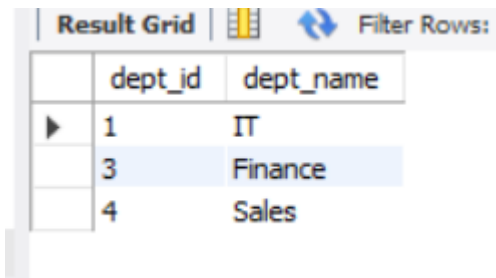
FROM employees e

WHERE e.dept_id = d.dept_id

AND e.salary <= 50000

);

Output



	dept_id	dept_name
▶	1	IT
	3	Finance
	4	Sales